

# More Logarithmic Equations and Properties of Logarithms (Homework)

Solve.

Answers

①  $8 \log_7 X + 5 = 21$

49

②  $-9 + \frac{\log_x 4}{6} = -3$

$\sqrt[36]{4}$

③  $-2 \log_x 9 + 4 = 0$

3

④  $\frac{\log_6 X}{5} + 13 = 13$

VII 7

⑤  $7 = 4 \log_5 X + 6$

IX  $\sqrt[4]{5}$

⑥  $8 = 3 - 2 \log_4 X$

VIII, X  $\frac{1}{32}$

\* Hint: Use rules VIII and X.

Convert to a sum or difference of logarithms.

⑦  $\log_7 (6X)$

$\log_7 6 + \log_7 X$

⑧  $\log_3 \left( \frac{5}{X} \right)$

$\log_3 5 - \log_3 X$

⑨  $\log_2 (ab)$

$\log_2 a + \log_2 b$

⑩  $\log_4 \left( \frac{c}{d} \right)$

$\log_4 c - \log_4 d$

Convert each sum or difference to one logarithm.

⑪  $\log_8 X + \log_8 62$

$\log_8 (62X)$

⑫  $\log_5 3 - \log_5 Y$

$\log_5 \left( \frac{3}{Y} \right)$

⑬  $\log_9 r + \log_9 m$

$\log_9 (rm)$

⑭  $\log_2 36 - \log_2 6$

$\log_2 6$

\* ⑮  $-\log_{15} c + \log_{15} b$

$\log_{15} \left( \frac{b}{c} \right)$

Convert each logarithm to a multiple of a logarithm.

⑯  $\log_6 2^4$

$4 \log_6 2$

⑰  $\log_5 3^8$

$8 \log_5 3$

⑱  $\log_2 b^7$

$7 \log_2 b$

⑲  $\log_x 9^{10}$

$10 \log_x 9$



Convert each expression from a multiple of a logarithm to a logarithm of a power.

$$(20) 7\log_3 4$$

$$\boxed{\log_3 4^7}$$

$$(21) 2\log_2 9$$

$$\boxed{\log_2 81}$$

$$(22) 5\log_3 X$$

$$\boxed{\log_3 X^5}$$

$$(23) 4\log_x 6$$

$$\boxed{\log_x 6^4}$$

Convert each logarithm to a sum or difference of multiples of logarithms. Write the Roman numeral of each logarithm property that you use.

$$(24) \log_9(4x^5)$$

$$\text{V, VII } \boxed{\log_9 4 + 5\log_9 X}$$

$$(25) \log_5\left(\frac{b^6}{2}\right)$$

$$\text{VI, VII } \boxed{6\log_5 b - \log_5 2}$$

$$(26) \log_3(c^2d)$$

$$\text{V, VII } \boxed{2\log_3 C + \log_3 d}$$

$$(27) \log_4\left(\frac{a^7}{b}\right)$$

$$\text{VI, VII } \boxed{7\log_4 a + \log_4 b}$$

$$(28) \log_6(x^8y^4)$$

$$\text{I, VII } \boxed{8\log_6 X + 4\log_6 Y}$$

$$(29) \log_2\left(\frac{c^3}{d^7}\right)$$

$$\text{VI, VII } \boxed{3\log_2 C - 7\log_2 d}$$

$$(30) \log_3(6x^5y^4)$$

$$\text{I, VII } \boxed{\log_3 6 + 5\log_3 X + 4\log_3 Y}$$

$$(31) \log_8\left(\frac{3b^4}{c^2}\right)$$

$$\text{VI, V, VII } \boxed{\log_8 3 + 4\log_8 b - 2\log_8 C}$$

Convert each expression to one logarithm.

$$(32) 4\log_3 M + \log_3 n$$

$$\text{VII, V } \boxed{\log_3 (M^4 n)}$$

$$(33) 2\log_6 2 - 3\log_6 C$$

$$\text{VII, VI } \boxed{\log_6 \left(\frac{4}{C^3}\right)}$$

$$(34) 3\log_4 2 + 7\log_4 X$$

$$\text{VII, V } \boxed{\log_4 (8X^7)}$$

$$(35) 8\log_5 X - 3\log_5 Y$$

$$\text{VII, VI } \boxed{\log_5 \left(\frac{X^8}{Y^3}\right)}$$

$$\star (36) -4\log_6 b + 2\log_6 C$$

$$\text{VII, VI } \boxed{\log_6 \left(\frac{C^2}{b^4}\right)}$$

$$(37) 2\log_3 7 + 5\log_3 X + 3\log_3 Y$$

$$\text{III, V } \boxed{\log_3 (49X^5Y^3)}$$

$$(38) 5\log_2 M + 2\log_2 n - 6\log_2 b$$

$$\text{VII, V, VI } \boxed{\log_2 \left(\frac{M^5 n^2}{b^6}\right)}$$

$$(39) 8\log_9 X - 4\log_9 Y + \log_9 7$$

$$\text{VII, VI, V } \boxed{\log_9 \left(\frac{7X^8}{Y^4}\right)}$$



If  $\log_k 7 \approx 3$  and  $\log_k 5 \approx 2.3$ , use this information along with Logarithm Properties V-VII to approximate the following values without a calculator. Write the Roman numeral of each logarithm property or exponent rule that you use.

40)  $\log_k 35$  V 5.3

41)  $\log_k 25$  V or VII 4.6

42)  $\log_k \sqrt{7}$  VII  $\frac{3}{2}$  or 1.5

43)  $\log_k \left(\frac{1}{7}\right)$  VII -3

44)  $\log_k \left(\frac{5}{49}\right)$  VI, VII or V -3.7